Advanced Machine Learning Project

Fashion Classifier AML

Automatic Clothing Category Recognition

Phase 1: Classical Machine Learning Evaluation

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Executive Summary

This project studies how well classical machine learning algorithms can classify clothing images. It uses Fashion-MNIST, a dataset of 70,000 grayscale images belonging to ten clothing categories. The project compares two ways to represent an image: normalized pixel values and Histogram of Oriented Gradients (HOG) features.

Four algorithms are evaluated: Logistic Regression, k-Nearest Neighbors (k-NN), Random Forest, and Linear Support Vector Machine (SVM). In the results recorded in the project, Random Forest using raw pixels achieves the highest classical-model test accuracy of 85.27%, with a macro F_1-score of 85.09%. The results also show that HOG helps the tested linear models but reduces Random Forest performance. Shirts are the most difficult category to recognize reliably.

The project also includes a Gradio interface for uploading an image, processing it for the model, and viewing a predicted category and category scores. This report focuses on Phase 1 classical machine learning; it does not present new Deep Learning experiments.

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1. Project Overview

1.1 Problem Statement

Clothing websites and retailers need to organize large numbers of product images. Manually assigning a category to every image takes time and may lead to inconsistent labels. This project aims to predict the category of a clothing image automatically.

The input is a 28×28 grayscale image. The output is one of ten labels, such as T-shirt/top, Trouser, Dress, Bag, or Ankle boot. The dataset has limited resolution and no colour information, so visually similar garments can be difficult to separate.

1.2 Objectives

- Explore the Fashion-MNIST dataset and its class labels.
- Compare raw-pixel features with HOG features.
- Evaluate four classical machine learning algorithms using a consistent training sample budget.
- Compare test performance and identify common classification errors.
- Provide an interactive interface for testing the trained model.

2. Dataset and Data Preparation

2.1 Fashion-MNIST

Fashion-MNIST contains 70,000 grayscale images of size 28×28: 60,000 official training images and 10,000 official test images. Each of the ten classes has 6,000 training images and 1,000 test images.

Table 1: Fashion-MNIST class labels

Label	Category	Label	Category
0	T-shirt/top	5	Sandal
1	Trouser	6	Shirt
2	Pullover	7	Sneaker
3	Dress	8	Bag
4	Coat	9	Ankle boot

2.2 Data Splitting

According to the project's documented experiment setup, the 60,000 official training images were divided into a 48,000-image training pool and a 12,000-image validation set. The classical models were trained using a shared subset of 12,000 images (1,200 per class). The separate official test set contains 10,000 images and is used for final evaluation. This separation helps reduce data leakage during model selection.

3. Feature Engineering and Algorithms

3.1 Image Features

Raw pixels: Each image is flattened from 28×28 pixels into 784 values. Pixel intensities are divided by 255 so they range from 0 to 1.

HOG features: Histogram of Oriented Gradients describes local edge directions and outlines. The project

configuration produces 324 HOG features per image. HOG can help a classifier focus on garment shape instead of each individual pixel value [2].

Table 2: Feature representations used in the project

Representation	Features	Simple explanation
Raw pixels	784	Uses the normalized intensity of every image pixel.
HOG	324	Summarizes local edge directions and shape information.

3.2 Classical Models

- **Logistic Regression:** Learns weighted combinations of features to estimate class scores. It is generally fast at prediction time.
- **k-Nearest Neighbors:** Predicts a category using the labels of similar training examples.
- **Random Forest:** Combines many decision trees to capture non-linear patterns in the image features [3].
- **Linear SVM:** Learns decision boundaries that separate the categories with a large margin [4].

4. Experimental Method

The project compares the four algorithms using both feature representations. Hyperparameters are selected using the validation set, and the reported final scores are measured on the official test set. The results below are taken from the project's documented benchmark outputs; this report does not claim that the experiments were rerun while preparing the document.

The main metrics are:

- **Accuracy:** The percentage of all images classified correctly.
- **Macro F_1-score:** The average F_1-score across the ten classes, giving each category equal weight.
- **Inference latency:** The reported time required to predict one image. It depends on the hardware and measurement setup.

5. Results and Comparison

Table 3 summarizes the recorded test-set results for the classical ML configurations. The highest test accuracy in this table is achieved by Random Forest using raw pixels.

Table 3: Documented test results for classical ML models

Model	Features	Accuracy	Macro F_1	Latency
Random Forest	Pixels	**85.27%**	**85.09%**	0.100 ms
Logistic Regression	HOG	83.27%	83.22%	0.094 ms
Linear SVM	HOG	82.56%	82.40%	0.094 ms
k-NN ($k = 5$)	Pixels	82.41%	82.40%	0.072 ms
Logistic Regression	Pixels	82.27%	82.22%	0.002 ms
Random Forest	HOG	81.97%	81.94%	0.189 ms
k-NN ($k = 5$)	HOG	81.09%	81.15%	0.137 ms
Linear SVM	Pixels	80.03%	79.84%	0.002 ms

5.1 Main Findings

- **Best classical model:** Random Forest with raw pixels records 85.27% test accuracy.
- **HOG and linear models:** HOG improves Linear SVM from 80.03% to 82.56% test accuracy and improves Logistic Regression from 82.27% to 83.27%.
- **HOG and Random Forest:** Random Forest accuracy falls from 85.27% with raw pixels to 81.97% with HOG features. The feature representation that helps one algorithm does not necessarily help every algorithm.
- **Speed:** Pixel Logistic Regression has the lowest reported latency in the table. These timings should be treated as hardware-dependent measurements, not universal guarantees.

6. Error Analysis

The documented per-class results show that Shirt is the most difficult category for the best classical model, with a reported recall of 52.9%. Trouser, Bag, Sandal, and Sneaker are recognized more reliably in the same results. A likely reason is that upper-body garments share similar outlines in small grayscale images. For example, a Shirt can resemble a T-shirt/top, Pullover, or Coat.

This error pattern suggests that overall accuracy alone does not show every weakness of a classifier. Reviewing class-level recall and a confusion matrix helps identify categories that need improvement.

7. Application and User Interface

The project includes a Gradio web application. A user can upload a clothing image, choose an image-background option, request a prediction, and view the predicted category and class scores. The interface also displays the processed 28×28 image so the user can see the input representation passed to the model.

The preprocessing pipeline converts the uploaded image to grayscale, resizes it while preserving its aspect ratio, and pads it to the required input size. Since Fashion-MNIST contains small, centered grayscale garments, predictions on ordinary photographs with people, complex backgrounds, or multiple objects may be less reliable.

8. Limitations and Future Work

8.1 Limitations

- The images are only 28×28 pixels and grayscale, so fine details and colour are unavailable.
- Similar-looking upper-body garments are often confused.
- Real-world photographs may contain backgrounds, poses, shadows, and multiple objects not represented in the same way in Fashion-MNIST.
- The reported latency depends on the hardware and the evaluation method.

8.2 Conclusion

This Phase 1 project compares four classical machine learning algorithms using raw pixels and HOG features. In the documented results, Random Forest with raw pixels achieves the strongest classical test accuracy at 85.27%. HOG improves the tested linear models but reduces Random Forest performance. The analysis also identifies Shirt as a difficult class and demonstrates how a Gradio interface can make the classifier easier to test. These results provide a clear baseline for understanding the strengths and limitations of classical ML on Fashion-MNIST.

References

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